

AIR-INSULATED RING MAIN UNIT

PRAL

Metal-enclosed switchgear for secondary distribution networks
12 kV & 24 kV



630 A

RATED BUSBAR
CURRENT

25 kA

SHORT-TIME
WITHSTAND · 1 S

SF6-free

AIR-INSULATED
SWITCHING

IEC

62271 SERIES
COMPLIANT

01 · COMPANY PROFILE

ENGINEERING THE FUTURE OF POWER WITH PRECISION AND INNOVATION

Since our inception in 2012, Powerline has been driven by a singular mission: to serve as the reliable backbone of the regional and Egyptian power sector. We are not merely a contracting and supply firm — we are a fully integrated engineering partner specialising in the design and manufacture of Medium Voltage (MV) and Low Voltage (LV) power distribution systems, adhering to the most stringent international technical standards.

A Legacy of Reliability

Built on foundations of commitment and precision, Powerline has established itself as a hallmark of quality in major national and industrial projects. Our track record reflects a consistent ability to overcome complex engineering challenges and deliver customised solutions for the modern electrical grid.

Solutions Beyond Expectations

Our MV switchgear, Ring Main Units and LV main and sub-main distribution boards are characterised by exceptional operational efficiency and a long life cycle. We bridge mechanical durability with digital intelligence, ready for integration with the latest smart-grid and control technologies.

Quality is not a department — it is our work culture



Production & Manufacturing

Advanced production lines ensure the highest accuracy in the assembly and calibration of MV & LV switchgear, in full compliance with IEC international standards.



Engineering & Design

Our design engineers develop layouts that prioritise maximum safety, operational continuity and seamless serviceability across the installation's life.



Testing & Quality Control

Every unit leaving our facilities undergoes rigorous routine testing to ensure durability under the harshest operating conditions — total peace of mind for our clients' investments.

2012

ESTABLISHED · CAIRO

2MANUFACTURING
FACILITIES**MV+LV**

INTEGRATED PORTFOLIO

IECCERTIFIED QUALITY
SYSTEM

" Powerline: Performance Excellence, Execution Precision & Sustainable Solutions. "

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How to use this catalogue

Sections 07–09 give assembly-level data. Sections 11–15 detail the switching devices and components. Sections 16–19 support unit selection, configuration and installation planning.

02 · PRODUCT OVERVIEW

PRAL AT A GLANCE

PRAL is a metal-enclosed, fully air-insulated Ring Main Unit for indoor secondary distribution at 12 kV and 24 kV. It is built around the globally proven ABB NAL / NALF three-position switch-disconnector — a completely SF6-free switching solution.

- Metal-enclosed unit for indoor installation, air-insulated system (AIS)
- Three-position function: **Closed – Open – Earthed**, with separate shafts for load and earth switching
- **SF6-free**: no gas to monitor, refill or recycle — simple, environmentally responsible technology
- All operations from the front; interlocking driven by the switch position protects the operator
- NALF switch-fuse combination with ABB CEF current-limiting fuses controls the full range of overload and short-circuit currents
- Fuse striker pin trips the mechanism automatically on fuse operation
- Maintenance-free, climate-independent and easy to install



PRAL ring main unit — on the Powerline factory floor

A key breaking element

PRAL is an attractive solution for enclosed switchgear and compact transformer substations, wherever a reliable, economical air-insulated RMU is required.

12 / 24 kV

RATED VOLTAGE

630 A

MAIN BUSBAR

25 kA

SHORT-TIME · 1 S

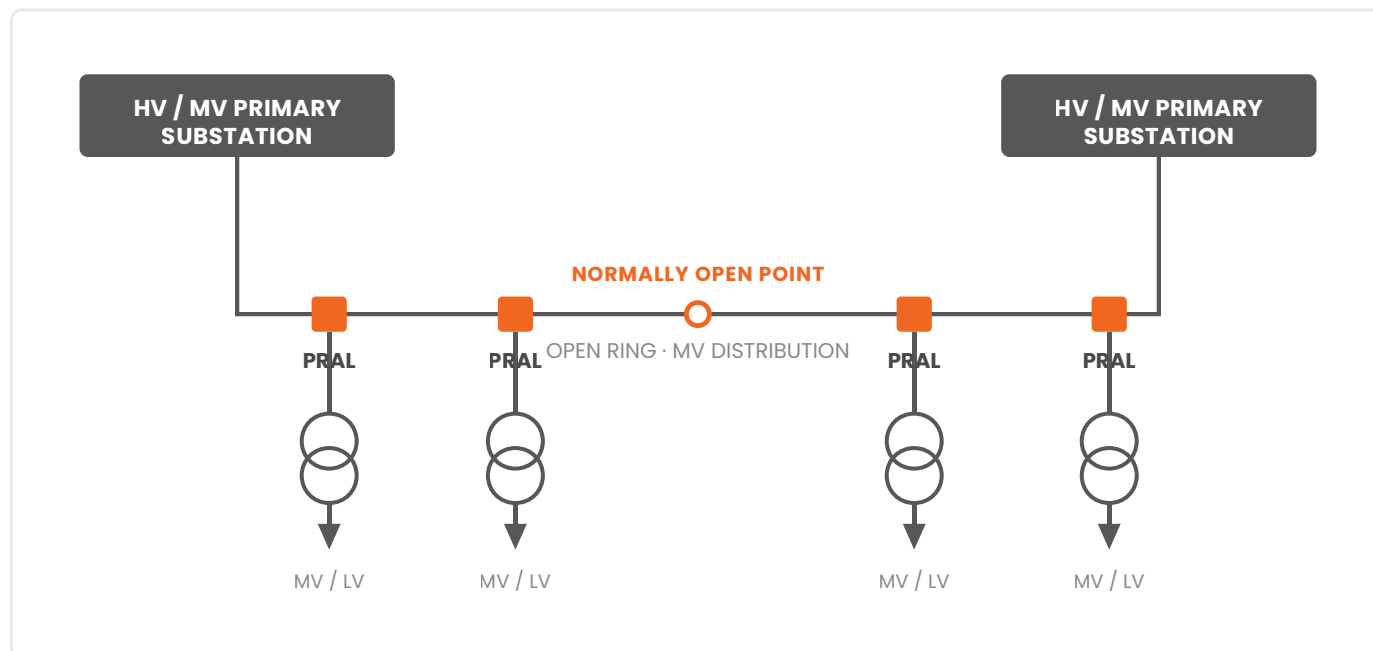
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SF6 · ZERO GAS
HANDLING

Designation. Functional units are combined to build the required scheme: **PCC** Powerline cable cubicle · **PFC** Powerline fuse cubicle · **PMC** metering cubicle. See Functional Units, page 20.

FIELD OF APPLICATION

PRAL connects, supplies and protects distribution transformers and cable feeders on open-ring or radial MV networks. The NAL/NALF switch-disconnector is suitable for all medium voltage primary and secondary distribution systems, making PRAL a universal building block for the modern grid.



Typical open-ring secondary distribution network with PRAL units at each MV/LV substation

Typical duties

- Ring main and radial network switching (incoming / outgoing feeders)
- Distribution transformer protection by NALF switch-fuse combination
- Line switching and network sectionalising
- Utility metering points with dedicated measurement cubicles

Typical environments

- Utility distribution networks
- Compact secondary substations (CSS)
- Industrial workshops & factories
- Solar & wind grid connection
- Residential & building applications
- Infrastructure projects
- Transformer substations

SAFETY BY DESIGN

Every switching operation on PRAL is designed to be safe for personnel — even with power present in the cables. Interlocks operate automatically according to the switch position, so the safe sequence is enforced by the equipment itself, not by procedure alone.



Safe with live cables

The unit can be approached and operated safely while cables are energised. Live parts are metal-enclosed and segregated from the operator.



Automatic interlocking

Interlocks driven by the switch position prevent any incorrect operating sequence. Operation is only possible with the door fully closed.



Earthed fuse access

The fuse compartment can only be accessed once the load break switch is in the earthed position — no exposure to live parts.



Clear status indication

A mimic diagram on the front fascia gives a trusted, unambiguous indication of the switching state — with independent position indication for switch and earthing switch.



Voltage detection

A capacitive voltage indicator (IEC 61958) fed from bushing-integrated capacitors confirms whether the connected cables are live.



Fail-safe fuse tripping

On fuse operation, the striker pin trips the switch mechanism automatically, disconnecting all three phases of the faulted circuit.

IAC · AF

Internal arc classification with front accessibility (AF) and IP42 protection on the front face: the operator standing at the front of the unit is protected. Enclosure finished in RAL 7035.

PRODUCT FEATURES



Technology

- Metal-enclosed unit for indoor installation
- Fully air-insulated system (AIS) with insulated busbars
- Three-position load break switch: Closed – Open – Earthed
- Proven ABB NAL / NALF switching technology
- Easy installation, independent of climate



Durability

- SF6-free: no gas system to monitor or maintain
- Robust modular switch: frame, insulators and current-carrying parts
- Arc quenching by combined air and gas blast – high switching capacity
- Voltage ratings tested with 100 operations at 630 A load current
- Front-operated with automatic position-based interlocking



Safety

- Safe operation with power present in cables
- Mimic diagram with clear operation status
- Operation only possible with the door closed
- Fuse compartment accessible only when earthed
- Voltage detector to verify live cables



Cost saving

- Maintenance-free – only HRC fuse replacement after operation
- Compact design – minimum space to install and operate
- Long service life with no replacement costs
- Economical air insulation with no end-of-life gas recycling
- Local manufacture: fast delivery and lifetime support

Compact by design

A complete 3-way ring main unit – two incoming ring switches and one transformer protection way – occupies a footprint of roughly **1.85 × 0.78 m** at 12 kV. Where space is the constraint, PRAL fits: substation rooms, kiosks and compact secondary substations.

654

MM · WAY
WIDTH (12 KV)

780

MM · DEPTH

1800

MM · HEIGHT

CONSTRUCTION & DESIGN

Each PRAL functional unit is a free-standing, metal-enclosed cubicle (LSC2A partition class) built around four clearly segregated zones. Cubicles bolt together into multi-way assemblies sharing a common insulated busbar system.

Compartment	Description
Busbar zone	Air-insulated 630 A main busbars, fully insulated for maximum reliability, running the full width of the assembly
Switching device	NAL / NALF three-position switch-disconnector with earthing switch; horizontal front cable bushings with integrated capacitive dividers for voltage indication
Cable compartment	Front-access termination for single-core cables up to 300 mm ² , voltage indication, earthing facilities and self-regulating anti-condensation heater
Fuse compartment (PFC)	Three HV HRC fuses with striker tripping; access interlocked — only possible with the way earthed on both sides
LV compartment	Secondary wiring, auxiliary devices, indication and control accessories (230 V AC), factory-wired to terminal blocks

Enclosure

- Sheet-steel enclosure, powder coated RAL 7035
- IP42 on the front face · internal arc classification AF
- Mimic diagram, padlocking and key-locking facilities
- Painted left & right end covers · lifting hooks
- Address labels on all functional units

Two pole-centre variants

12 kV units are built on the NAL/ NALF PC170 (170 mm pole distance); 24 kV units use the PC275 (275 mm) — the air-insulation clearances that give each voltage class its dimensions and its generous dielectric margins.

Safe segregation — LSC2A

Loss-of-service-continuity class LSC2A: the cable compartment of one way can be opened while the busbar and the other functional units of the assembly remain energised and in service.

IP42

FRONT-FACE
PROTECTION

IAC · AF

INTERNAL ARC
CLASSIFIED

630 A

INSULATED BUSBAR
SYSTEM

RAL 7035

POWDER-COAT FINISH

MAIN CHARACTERISTICS

Ratings of the complete PRAL assembly. Component-level data for the NAL / NALF switch-disconnector is given on pages 14–15.

Characteristic	Unit	PRAL 12	PRAL 24
Rated voltage	kV	12	24
Service voltage	kV	11	22
Rated frequency	Hz	50	50
Rated power-frequency withstand voltage (1 min)	kV	28	50
Rated lightning impulse withstand voltage (1.2/50 µs)	kV	75	125
Rated current, main busbars	A	630	630
Rated short-time withstand current (1 s)	kA	25	25
Rated peak withstand current	kA	62.5	62.5
Internal arc classification	—	IAC · AF · front accessibility	
Degree of protection, front face	—	IP42	
Partition class	—	LSC2A	
Insulation	—	Air (AIS) — SF6-free	
Installation	—	Indoor	
Enclosure colour	—	RAL 7035	

Voltage detection

Capacitive voltage presence indication system (VPIS) to IEC 61958, supplied by bushing-integrated capacitive dividers.

Earth fault indication

Earth fault indicator supplied on ring switch cubicles for fast localisation of cable faults.

Cable connection

Front cable termination, single-core cables up to 300 mm² (switch ways) / 95 mm² (fuse ways).

STANDARDS & SERVICE CONDITIONS

PRAL is designed, manufactured and routine-tested in accordance with the IEC 62271 series.

Standard	Scope
IEC 62271-1	High-voltage switchgear and controlgear — Part 1: Common specifications
IEC 62271-100	Part 100: Alternating-current circuit-breakers
IEC 62271-102	Part 102: Alternating-current disconnectors and earthing switches
IEC 62271-103	Part 103: Switches for rated voltages above 1 kV up to and including 52 kV
IEC 62271-105	Part 105: Alternating-current switch-fuse combinations
IEC 62271-200	Part 200: AC metal-enclosed switchgear and controlgear for rated voltages above 1 kV up to and including 52 kV
IEC 61958	Voltage presence indicating systems (VPIS)

Environmental & additional data

Condition	Specification
Operating temperature	-5 °C to +40 °C
Storage temperature	-5 °C minimum
Installation altitude	Up to 1,000 m above sea level
Relative humidity	Max. 95 %
Protection index	IP42 on front face
Internal arc classification	AF — front accessibility
Switchgear colour	RAL 7035
Busbar treatment	Insulated
Packing	Domestic
Factory acceptance test	Powerline standard FAT (one day)

Why it matters

- **Type-conformant design** — ratings verified against the IEC 62271 series requirements for metal-enclosed switchgear.
- **Routine tests on every unit** — dielectric, mechanical operation and auxiliary circuit checks before dispatch.
- **Witnessed FAT** — a structured one-day factory acceptance test programme open to client witnessing.

The NAL/NALF switching device itself is qualified for -40...+40 °C ambient — giving comfortable design margin under Egyptian and regional service conditions. For installations above 1,000 m, contact Powerline engineering.

CONTROL, WIRING & STANDARD EQUIPMENT

Item	Specification
Local control voltage	230 V AC
Auxiliary voltage, ES electromechanical lock	230 V AC
Control circuit cross-section	1 mm ²
Voltage circuit cross-section	1.5 mm ²
Current circuit cross-section	2.5 mm ²
Earthing circuit cross-section	2.5 mm ²
Interconnection circuit cross-section	2.5 mm ²
Wiring cable type	Standard PVC

Wiring colour code	Colour
Auxiliary AC circuits	Black
Auxiliary DC circuits	Black
CT circuits	Black
VT circuits	Black
Earthing circuits	Yellow / Green

Fitted as standard

Equipment	Supply
Internal lighting	Yes
Anti-condensation heater, self-regulating type	Cable compartment
Voltage presence indicating system	Fixed lamps, type VPIS
Address labels on functional units	Yes
Operating handle for switch-disconnector	1
Lifting hooks	1 set
Painted end covers, left & right	1 + 1

Smart-grid ready

With auxiliary contacts, shunt trip options and fault passage indication, PRAL integrates readily with RTUs and SCADA for remote monitoring and network automation.

Switching devices

At the heart of PRAL is the ABB NAL / NALF — a medium voltage indoor switch-disconnector well known around the globe. Our cubicle design fits the NAL/NALF arc-quenching system, enabling high switching capacity with fully air-insulated technology.

ABB NAL · ring & network ways

Three-position switch-disconnector with earthing switch — pages 13–14

ABB NALF · transformer protection

Switch-fuse combination with CEF current-limiting fuses — page 15



ABB NAL THREE-POSITION SWITCH-DISCONNECTOR WITH EARTHING SWITCH

3

POSITIONS: C / O / E

AIR

SF6-FREE QUENCHING

E3 · C2

ENDURANCE CLASSES

100×

OPERATIONS AT 630 A

ABB NAL / NALF SWITCH-DISCONNECTOR

The NAL/NALF is a globally proven medium voltage indoor switch-disconnector. NAL switches load currents and small fault currents; combined with a current-limiting fuse it becomes the NALF — controlling the full range of overload and short-circuit currents.

Arc quenching — two combined effects

- **Current-independent air blast** generated by a piston as the contacts open — reliable interruption of small currents
- **Current-dependent gas blast** from the arc chute walls, increasing with current — powerful quenching at high currents
- The balanced combination gives high switching reliability across the whole current range, verified by 100 switching operations at 630 A

Modular by design

- Basic unit: frame with insulators and current-carrying parts
- Operating mechanisms: snap-action type **K** or stored spring-energy type **A**
- Earthing switch type E with making capacity; two separate shafts for load and earth functions
- NALF: integrated fuse base with striker-pin tripping and blown-fuse indication

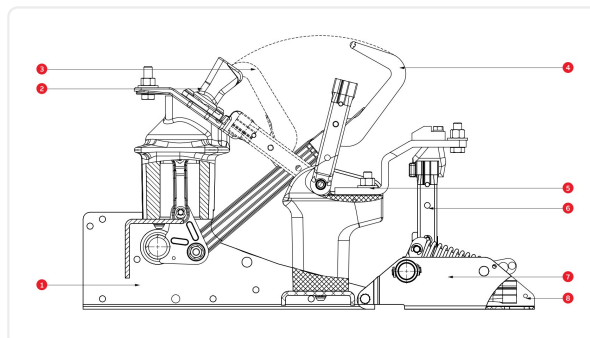
Available versions

NAL — line switch-disconnector

NALF — switch-fuse combination

NALFO — opening-site fuse base

NAL-H / NALF-H / NALFO-H — harsh operating conditions



1	Switch-disconnector	5	Pivot side
2	Opening side	6	Closed position (ES)
3	Closed position	7	Earthing switch
4	Open position	8	Open position (ES)

NAL switch-disconnector with earthing switch type E

Cable bushing with built-in sensing

Horizontal front cable bushings carry integrated capacitors for the voltage indication system — live-cable status is always visible at the fascia.

Accessories

Auxiliary contacts, shunt trip, undervoltage coil, key locks and coil preventing lever operation in the earth position.

NAL ELECTRICAL CHARACTERISTICS

Switch-disconnector type NAL, 630 A class, according to IEC 62271-103.

Characteristic	Unit	NAL 12 (PC170)	NAL 24 (PC275)
Rated voltage	kV	12	24
Rated current	A	630	630
Rated frequency	Hz	50	50
Power-frequency withstand voltage — common / across isolating distance	kV	28 / 32	50 / 60
Lightning impulse withstand voltage — common / across isolating distance	kV	75 / 85	125 / 145
Rated short-circuit making current	kAp	67	50
Rated peak withstand current	kAp	82	82
Rated short-time withstand current (1 s / 2 s)	kA	31.5 / 25	31.5 / 25
Rated mainly active load-breaking current	A	630	630
Rated cable / line-charging breaking current	A	150	80
Rated earth-fault breaking current	A	150	75
Opening time	ms	40 – 60	
Pole distance	mm	170	275
Ambient air temperature (device)	°C	–40 ... +40	
Endurance / switching classes	—	M1 · E3 · C2	

Data of the switching device according to the manufacturer's type tests (ABB NAL/NALF catalogue). Ratings of the complete PRAL functional unit are given on page 09.

Type K — snap-action mechanism

Spring-assisted opening and closing at speeds independent of the operator — the standard choice for ring and network switching ways (PCC).

Type A — stored spring-energy mechanism

Stored-energy operation enabling tripping by fuse striker pin, shunt trip and undervoltage coil — required for NALF switch-fuse combinations (PFC).

NALF & EARTHING SWITCH CHARACTERISTICS

NALF switch-fuse combination — IEC 62271-105

Characteristic	Unit	NALF 12	NALF 24
Rated voltage	kV	12	24
Max. rated current of fuse-link (CEF)	A	125	80
Rated short-circuit breaking current	kA	63	—
Max. breaking capacity in co-operation with fuses	A	1,600	900
Rated take-over current	A	1,450	—
Opening time	ms	40 – 60	
Pole distance	mm	170	275

Values valid for the ABB CEF current-limiting fuse-link family. Other fuse-links may be used where IEC 62271-105 co-ordination requirements are met.

Earthing switch type E — IEC 62271-102

Characteristic	Unit	12 kV	24 kV
Rated peak withstand current	kAp	82	82 ⁽¹⁾
Rated short-time withstand current (1 s)	kA	31.5	31.5
Rated short-circuit making current	kAp	67	50 ⁽¹⁾
Power-frequency withstand voltage	kV	28	50
Lightning impulse withstand voltage	kV	75	125
Making capability class	—	E2	

(1) When fed from the switch-disconnector side.

Downstream earthing on fuse ways

PFC cubicles additionally carry a downstream earthing switch (limited making capacity, 2 kA) so the fuse compartment is always accessed with both sides earthed.

67 kAp

MAKING CAPACITY · 12
KV

82 kAp

PEAK WITHSTAND

1,600 A

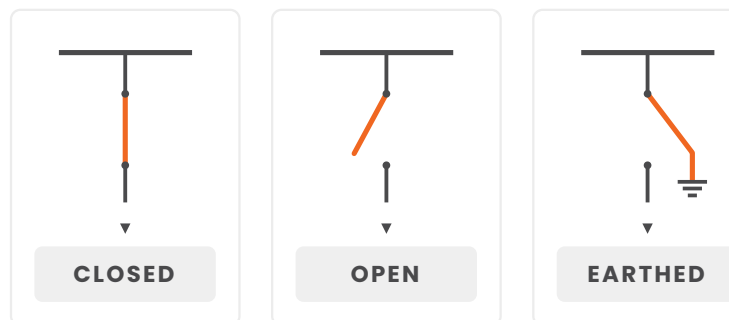
BREAKING WITH FUSES ·
12 KV

E2

ES MAKING CLASS

SWITCHING POSITIONS

Three positions, one truth



- Three-position load break switch with disconnecter and earthing switch in one device
- Operating mechanism with **two separate shafts** for the load and earth functions
- Independent switch position indication for LBS and earthing switch
- Transition line ↔ earth always passes through **Open**
- Padlock and key-lock provisions on the operating seats

Standard features

Feature

Three-position load break switch with disconnecter and earthing switch

Operation mechanism with two separate shafts for load and earth function

Switch position indication for LBS and ES

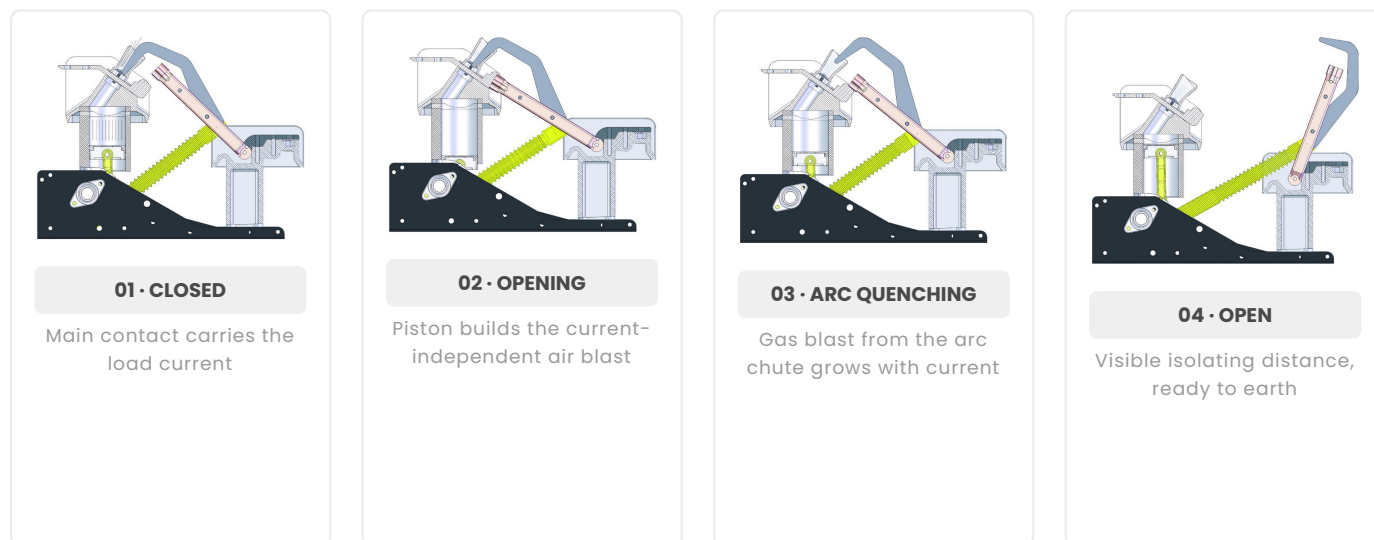
Cable bushing horizontal in front, with integrated capacitor for voltage indication

Voltage indicating system (VPIS)

Operator-independent switching

Snap-action (K) and stored-energy (A) mechanisms open and close the switch at speeds independent of the operator – consistent, repeatable switching every time.

Arc interruption sequence — inside the NAL interrupter



INSTRUMENT TRANSFORMERS

Current transformers

Ring-core type for protection and indication on cable ways; cast-epoxy DIN types for the metering cubicle.

Characteristic	Unit	Value
Max. system voltage	kV	0.6
Primary current	A	7.2 – 230.4
Secondary current	A	0.075
Rated burden	VA	0.1
Accuracy class	—	10P80
Short-time current	kA / 1 s	20
Rated frequency	Hz	50 / 60

Metering duty (PMC): cast-epoxy DIN CTs, primary per client requirement, core 1 A · 10 VA · class 0.5 · FS5 · 50 Hz.

Voltage transformers

Single-pole (phase-to-earth) epoxy-resin VTs for measurement and metering, with or without protection fuse.

Characteristic	Unit	Value
Highest voltage for equipment	kV	3.6 – 40.5
Primary voltage	V	3000/√3 – 35000/√3
Secondary voltage	V	100/√3 – 120/√3 · 100/3 – 120/3
Rated frequency	Hz	50
Fusing	—	With / without fuse

Metering duty (PMC): phase-earth VTs, core Us = 0.11/√3 kV · 50–100 VA · class 0.5.

Selection note. Instrument transformer ratios and burdens are engineered per project — primary currents, cores and accuracy classes are matched to the utility metering code and protection scheme. Powerline adjusts burdens accordingly at order stage.

OPTIONS & ACCESSORIES

Every PRAL way can be tailored with supervision, comfort and safety accessories — factory fitted and tested.



Voltage presence indicator (VPIS)

Push-button LED voltage indicator to IEC 61958, powered by bushing-integrated capacitive dividers, confirming whether cables are live.



Earth fault indicator

Fault passage indication on ring ways for rapid localisation of cable faults and shorter restoration times.



Self-regulating heater

Anti-condensation heater of self-regulating type in the cable compartment — no thermostat to fail, no over-heating risk.



Internal lighting

Cable compartment illumination for safe termination, inspection and maintenance work — fitted as standard.



Locking schemes

Padlocks, key locks and 230 V AC electromechanical locking on the earthing switch mechanism for switching-programme enforcement.



Trip & release options

Shunt trip and undervoltage coil options on stored-energy mechanisms; coil preventing lever operation in the earth position.



Auxiliary contacts

Position signalling of switch and earthing switch for interlocking, SCADA and remote indication.



Fuse status signalling

Blown-fuse indicator and striker-driven tripping on switch-fuse ways, disconnecting all three phases.



Handling & finishing

Lifting hooks, painted left and right end covers, and address labels on all functional units — delivered ready to install.

HRC POWER FUSES

PRAL switch-fuse ways (PFC) carry three HV HRC current-limiting fuses. The HRC power fuse interrupts high fault currents **before the first peak**, dramatically reducing the withstand duty imposed on the protected transformer and the associated equipment.

- Current-limiting interruption — cuts the fault before peak value
- High breaking capacity in a compact, enclosed design suitable for in-panel use
- Striker pin operates the trip indicator and trips the switch mechanism, disconnecting all three phases
- Standard equipment: 24 kV fuses up to 80 A (per project selection)
- Combination behaviour per IEC 62271-105 with ABB CEF fuse-links

Coordination made simple

Fuse-link ratings below follow the fuse manufacturer’s selection tables for distribution transformers without secondary-side coordination constraints. Final selection considers inrush, cyclic loading and utility rules — Powerline engineering validates every order. Maximum fuse rating accepted by the switch: 125 A at 12 kV, 80 A at 24 kV.

Recommended fuse-link ratings [A] vs. transformer rating [kVA]

Service voltage	25	50	75	100	125	160	200	250	315	400	500	630	800	1000	1250	1600
11 / 12 kV	6	6	10	16	16	16	20	20	25	40	40	50	63	80	100	125
22 / 24 kV	6	6	6	6	10	10	16	16	16	20	20	25	40	40	50	63

Indicative values based on the fuse manufacturer’s tables (CEF-type, back-up class). Confirm final selection with Powerline engineering per IEC 62271-105 and the specific fuse type-test data.

< ½ cycle

CURRENT-LIMITING ACTION

80 A

STANDARD MAX. FITTED RATING

125 A

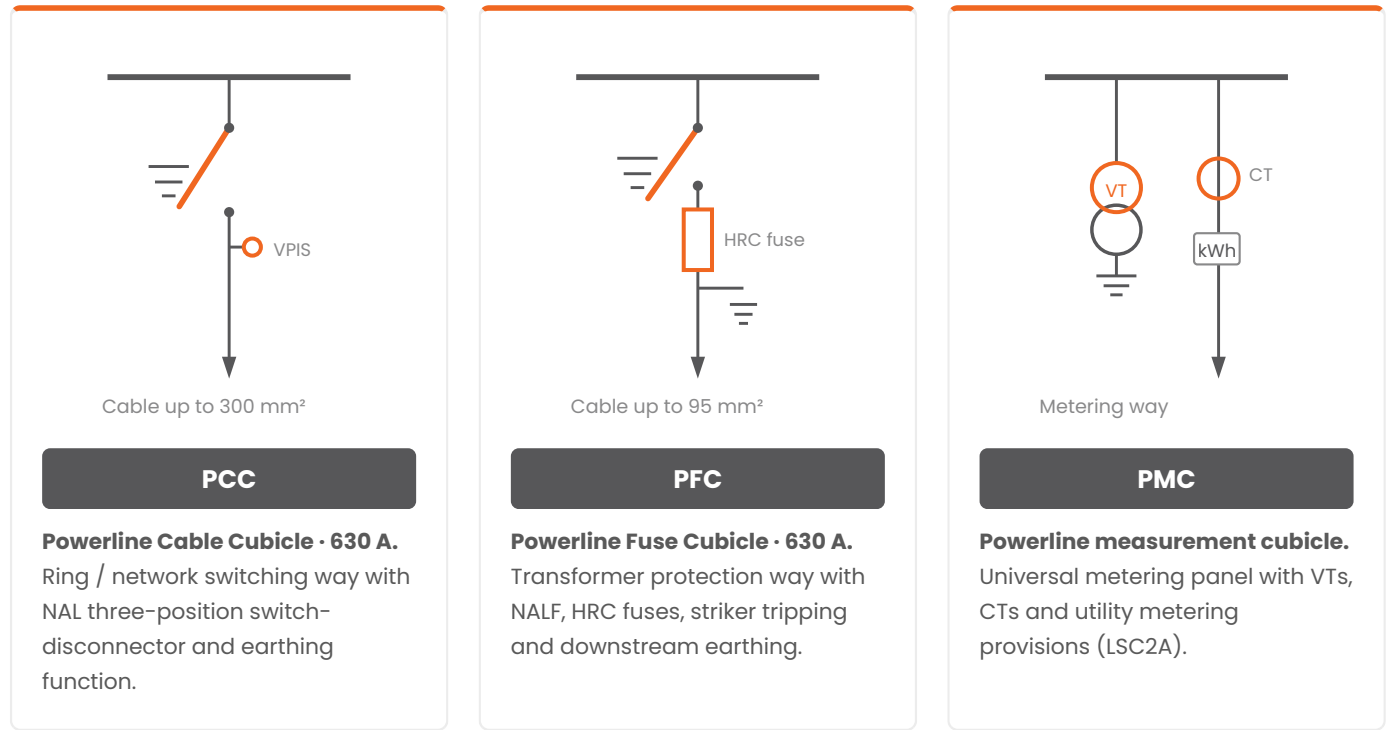
MAX. AT 12 KV (CEF)

3 φ

TRIPPING VIA STRIKER PIN

FUNCTIONAL UNITS

Three standardised cubicles combine into any secondary distribution scheme – from a single switch way to multi-way substations with transformer protection and utility metering.



Model designation



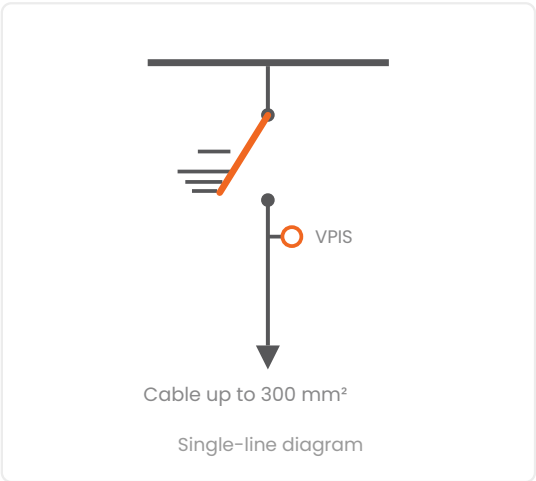
N = number of network LBS ways (PCC) · **F** = number of switch-fuse ways (PFC) · **M** = metering cubicle (PMC).

Example: **N2F1M** – two ring switch ways + one transformer protection way + metering: the classic compact substation.

PCC · POWERLINE CABLE CUBICLE

Ring and network switching way, rated 630 A. Metal-enclosed cubicle, LSC2A, hosting the NAL three-position switch-disconnector with earthing switch.

Qty	Composition
1	Air-insulated LBS cubicle, type LSC2A — load break switch-disconnector unit
1	NAL three-position switch-disconnector with earthing switch and single-spring (snap-action) operating device
1	Cable entry for 1-phase (single-core) cables up to 300 mm²
1	Voltage presence indicator with fixed lamps, type VPIS
1	Earth fault indicator
1	Anti-condensation heater, self-regulating, cable compartment
1	Internal light
1	Insulated busbar system
1	Low-voltage compartment



Dimensions	12 kV	24 kV
Width	654 mm	854 mm
Depth	780 mm	780 mm
Height	1,800 mm	1,800 mm

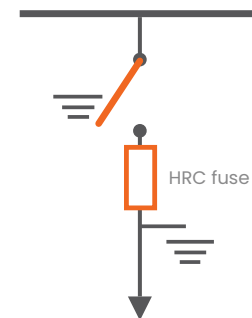
Options: key locking schemes, auxiliary contacts, electromechanical ES lock, earth fault indicator variants.

16 · FUNCTIONAL UNITS

PFC · POWERLINE FUSE CUBICLE

Transformer protection way, rated 630 A. NALF switch-fuse combination per IEC 62271-105 with automatic three-phase tripping on fuse operation and earthed-access fuse compartment.

Qty	Composition
1	Air-insulated LBS-F cubicle, type LSC2A — load break switch with fuse combination unit
1	NALF three-position switch-disconnector with earthing switch and stored-energy (2-spring) operating device
3	HV HRC fuses, 24 kV, up to 80 A (P&C or equivalent), with striker pin
1	Fuse base and holder · fuse tripping system · fuse status indication
1	Downstream earthing switch with limited making capacity (2 kA)
1	Cable entry for 1-phase (single-core) cables up to 95 mm ²
1	Voltage presence indicator with fixed lamps, type VPIS
1	Operating handle for switch-disconnector and earthing switch
1	Anti-condensation heater, self-regulating · internal light
1	Insulated busbar system · low-voltage compartment

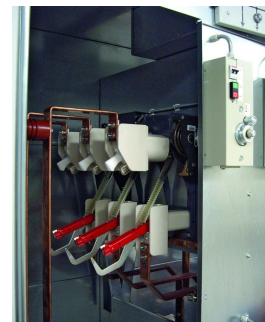


Cable up to 95 mm²

Single-line diagram

Dimensions	12 kV	24 kV
Width	654 mm	854 mm
Depth	780 mm	1,100 mm
Height	1,800 mm	2,050 mm

Fuse compartment access only with the way earthed — enforced by the interlocking system. At 24 kV the fuse cubicle is deeper and taller to house the 275 mm pole-centre NALF with fuses.

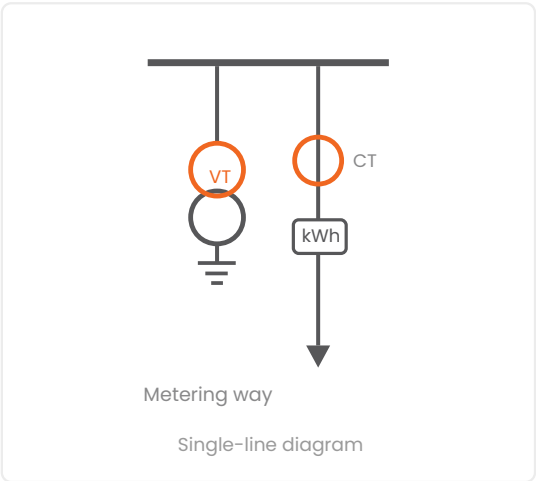


NALF with CEF fuses — fuse compartment

PMC · MEASUREMENT CUBICLE

Universal metering panel for utility revenue metering and network measurement, fully integrated into the PRAL busbar system.

Qty	Composition
1	Universal metering panel, cubicle type LSC2A
3	Cast-epoxy DIN current transformers — primary current per client requirement; core 1 A · 10 VA · class 0.5 · FS5 · 50 Hz
3	Voltage transformers, phase-to-earth — core $U_s = 0.11/\sqrt{3}$ kV · 50–100 VA · class 0.5
1	Voltage presence indicator with fixed lamps, type VPIS
1	Sealable door lock (utility sealing)
1	Insulated busbar system
1	Low-voltage compartment



Dimensions	12 kV	24 kV
Width (module, in assembly)	+ 780 mm	+ 780 mm
Depth (assembly)	780 mm	1,100 mm
Height (assembly)	1,863 mm	2,050 mm

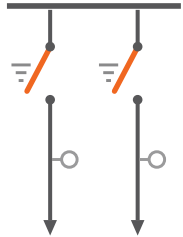
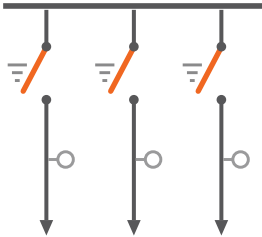
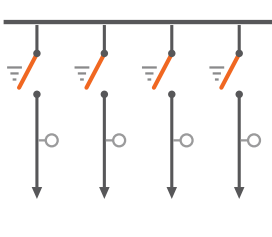
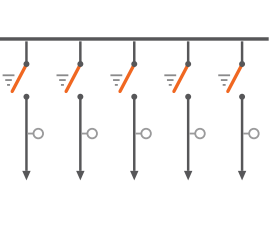
CT ratios and VT burdens are adjusted to the metering authority’s requirements at order stage.

17 · TYPES & DIAGRAMS

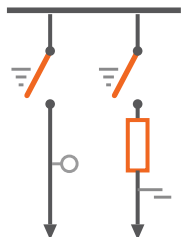
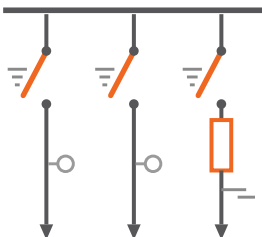
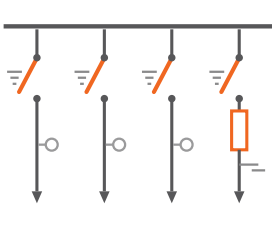
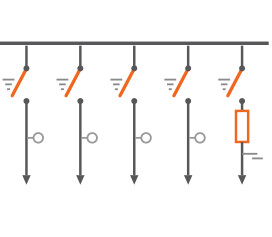
TYPES & DIAGRAMS

Standard PRAL assemblies with single-line diagrams. Model designation: **N** network switch way (PCC) · **F** switch-fuse way (PFC) · **M** metering cubicle (PMC). Dimensions W × H × D in mm, per voltage class.

1 Load break switch combinations — without metering

N1  12 kV · 654 × 1800 × 780 24 kV · 854 × 1800 × 780	N2  12 kV · 1254 × 1800 × 780 24 kV · 1654 × 1800 × 780	N3  12 kV · 1854 × 1800 × 780 24 kV · 2454 × 1800 × 780	N4  12 kV · 2454 × 1800 × 780 24 kV · 3254 × 1800 × 780	N5  12 kV · 3054 × 1800 × 780 24 kV · 4054 × 1800 × 780
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2 Transformer protection by fuses — without metering

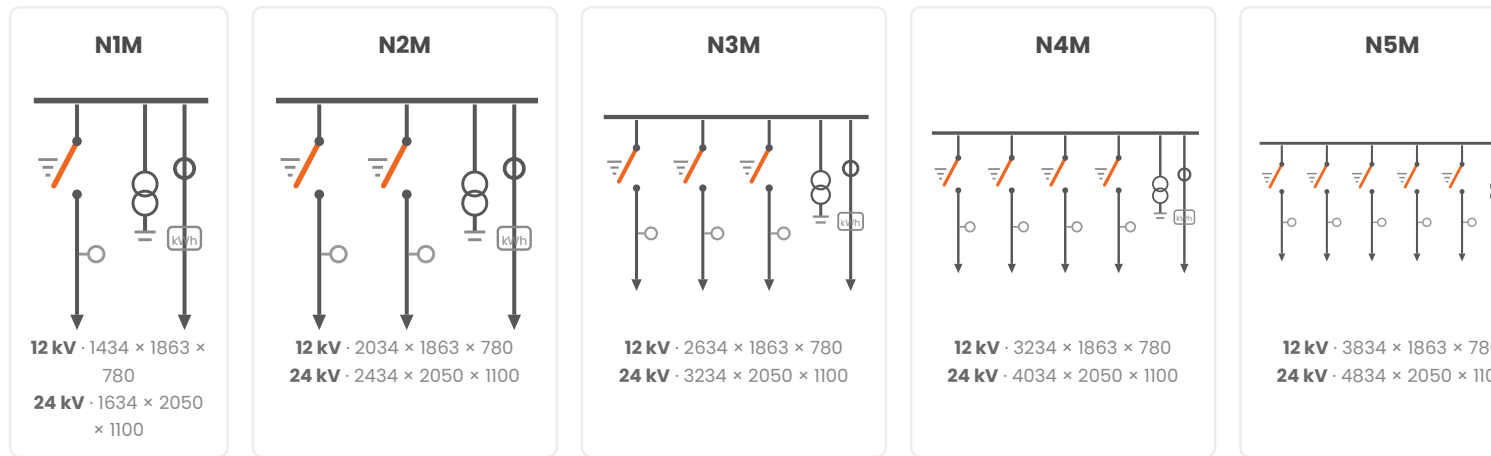
F1  12 kV · 654 × 1800 × 780 24 kV · 854 × 2050 × 1100	N1F1  12 kV · 1254 × 1800 × 780 24 kV · 1654 × 2050 × 1100	N2F1  12 kV · 1854 × 1800 × 780 24 kV · 2454 × 2050 × 1100	N3F1  12 kV · 2454 × 1800 × 780 24 kV · 3254 × 2050 × 1100	N4F1  12 kV · 3054 × 1800 × 780 24 kV · 4054 × 2050 × 1100
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Width rule

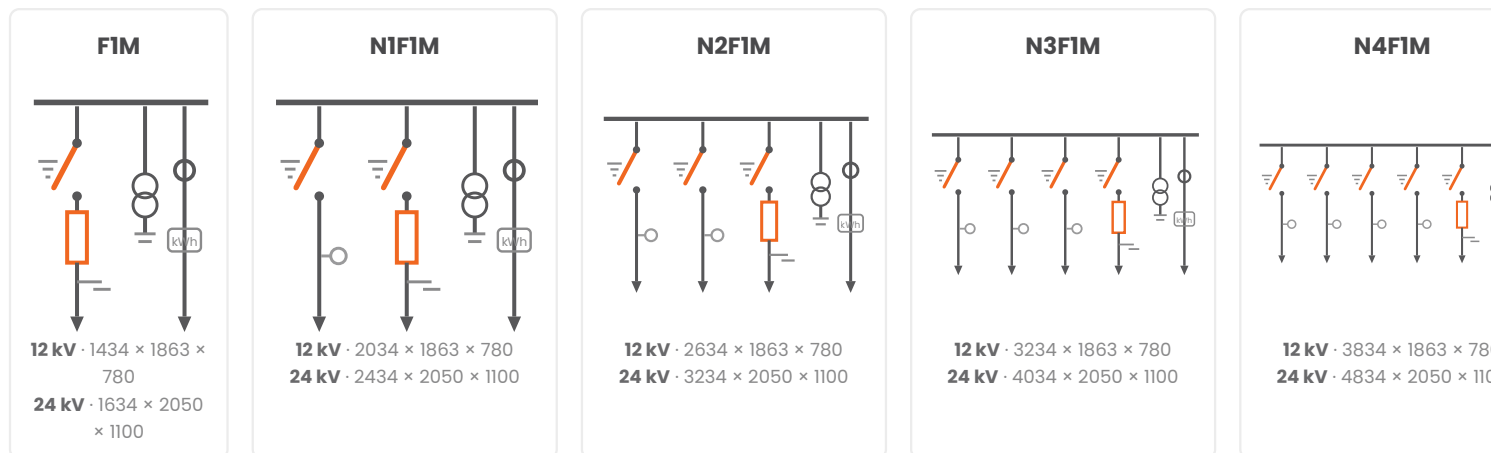
12 kV: 654 mm first way + 600 mm per additional way + 780 mm metering module. **24 kV:** 854 mm first way + 800 mm per additional way + 780 mm metering module. Fuse and metering combinations at 24 kV: height 2,050 mm · depth 1,100 mm.

COMBINATIONS WITH METERING

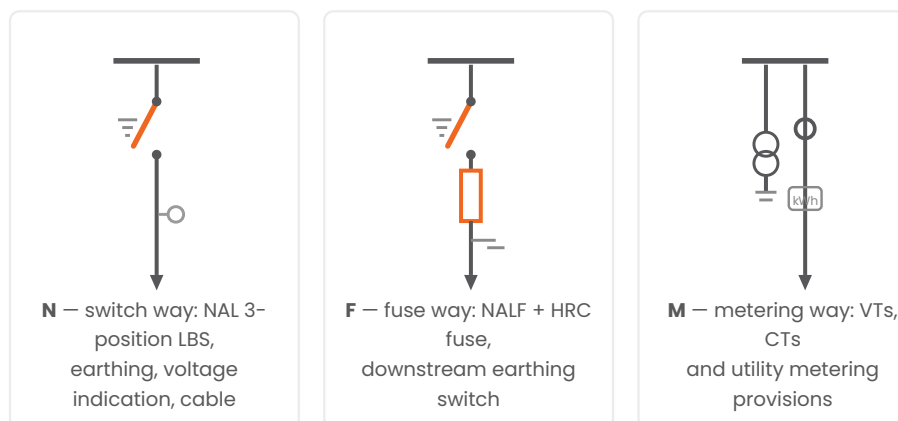
3 Load break switch combinations — with metering



4 Transformer protection by fuses — with metering



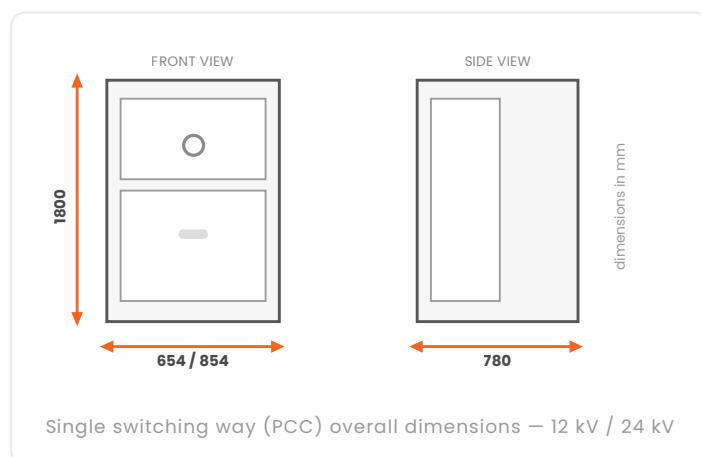
Symbol legend



Beyond the standard matrix

Larger schemes — more switching ways, multi-fuse and multi-metering arrangements — follow the same width rules. Every PRAL is assembled to the project single-line diagram; contact Powerline engineering for non-standard combinations.

INSTALLATION & SITE REQUIREMENTS



Module	12 kV	24 kV
First switching way, width	654 mm	854 mm
Additional way (in assembly)	+ 600 mm	+ 800 mm
PMC metering module (in assembly)	+ 780 mm	+ 780 mm
Height – switch assemblies	1,800 mm	1,800 mm
Height – fuse / metering assemblies	1,800 / 1,863 mm	2,050 mm
Depth – switch / fuse & metering	780 mm	780 / 1,100 mm

- **Indoor installation** on a level floor or plinth; the compact footprint suits substation rooms, kiosks and compact secondary substations
- **Front access only** required for operation, cable termination and fuse replacement – units can stand against the wall
- **Cable connection from the front**, single-core cables up to 300 mm² (switch ways) / 95 mm² (fuse ways); cable trench or duct below the unit recommended
- **Bolted assembly** of cubicles into multi-way units on site, or delivered as a complete factory-assembled RMU
- **Earthing** of the enclosure and earthing circuits per utility practice (2.5 mm² minimum earthing circuit section)
- **Easy handling** – lifting hooks fitted; painted end covers complete the assembly

Commissioning & life

After cable termination and functional checks, PRAL is ready for service. No scheduled maintenance is required – only HRC fuse replacement after a fuse operation. With air insulation there is no gas system to check, ever.

Service conditions: -5...+40 °C · humidity max. 95 %
· altitude up to 1,000 m (higher on request).

QUALITY, TESTING & CERTIFICATIONS

Routine tests — every unit

- Visual inspection, dimensional and build verification
- Power-frequency voltage test on the main circuit
- Voltage test on auxiliary and control circuits
- Mechanical operating tests and interlock verification
- Verification of correct wiring, indication and accessories
- Documented factory acceptance test (FAT) — one day, open to client witnessing

Type verification & approvals

- PRAL 12 kV & 24 kV boards approved and registered by the Egyptian electricity sector authorities (Egyptian Electricity Holding Company vendor accreditation), covering NAL 12 / NALF 12 (PC170) and NAL 24 / NALF 24 (PC275) equipment
- NAL / NALF switch-disconnectors and CEF fuse-links type-tested by ABB per IEC 62271-103 / -105 / -102
- Assembly designed and routine-tested per IEC 62271-200



National approval certificate — PRAL 12 & 24 kV with ABB NAL/NALF

Documentation with every delivery

Routine test certificates, as-built drawings, single-line diagram, operating manual and type-test evidence are supplied with each PRAL assembly.



Performance excellence. Execution precision. Sustainable solutions.

From engineering and manufacture to testing, delivery and lifetime support — Powerline is your integrated partner for MV & LV power distribution in Egypt and the region.

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For technical enquiries or quotations, please get in touch with our team — we respond with engineered solutions, not just prices.

12 / 24

KV RATED VOLTAGE

630 A

MAIN BUSBAR

SF6-free

AIR INSULATION

N · F · M

MODULAR FUNCTIONAL
UNITS